Terraform Task

Task Description:

Launch Linux EC2 instances in two regions using a single Terraform file.

AWS CLI and Terraform installations:

```
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_17$ aws --version
aws-cli/2.34.2 Python/3.13.11 Linux/6.6.87.2-microsoft-standard-WSL2 exe/x86_64.ubuntu.24
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_17$
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_17$ terraform --version
Terraform v1.14.3
on linux_amd64
+ provider registry.terraform.io/hashicorp/aws v6.37.0

Your version of Terraform is out of date! The latest version
is 1.14.8. You can update by downloading from https://developer.hashicorp.com/terraform/install
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_17$
```

Creating a main.tf file to launch 2 EC2 instances each in different regions [ap-south-1 and ap-south-2]:

```
GHOST@GHOST: ~/Projects/[

provider "aws" {
  alias  = "mumbai"
  region = "ap-south-1"
}

provider "aws" {
  alias  = "hyderabad"
  region = "ap-south-2"
}

# Choosing the OS Type to use

data "aws_ami" "ubuntu_mumbai" {
  provider    = aws.mumbai
  most_recent = true

  owners = ["099720109477"] # Canonical

  filter {
    name   = "name"
    values = ["ubuntu/images/hvm-ssd/ubuntu-jammy-22.04-amd64-server-*"]
  }
}

data "aws_ami" "ubuntu_hyderabad" {
  provider    = aws.hyderabad
  most_recent = true

  owners = ["099720109477"] # Canonical

  filter {
    name   = "name"
    values = ["ubuntu/images/hvm-ssd/ubuntu-jammy-22.04-amd64-server-*"]
  }
}

# EC2 Instance Creation

resource "aws_instance" "mumbai_ec2" {
  provider      = aws.mumbai
  ami           = data.aws_ami.ubuntu_mumbai.id
  instance_type = "t3.micro"

  tags = {
    Name = "Ubuntu-Mumbai"
  }
}

resource "aws_instance" "hyderabad_ec2" {
  provider      = aws.hyderabad
  ami           = data.aws_ami.ubuntu_hyderabad.id
  instance_type = "t3.micro"

  tags = {
    Name = "Ubuntu-Hyderabad"
  }
}

# Outputs

output "mumbai_public_ip" {
  value = aws_instance.mumbai_ec2.public_ip
}

output "hyderabad_public_ip" {
  value = aws_instance.hyderabad_ec2.public_ip
}
```

Initiating the terraform script:

```
GHOST@GHOST: ~/Projects/I

GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_17$ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.37.0...
- Installed hashicorp/aws v6.37.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_17$ terraform plan
data.aws_ami.ubuntu_mumbai: Reading...
data.aws_ami.ubuntu_hyderabad: Reading...
data.aws_ami.ubuntu_mumbai: Read complete after 1s [id=ami-095cc90aa5ddff518]
data.aws_ami.ubuntu_hyderabad: Read complete after 1s [id=ami-04d1c5d5066b58e37]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.hyderabad_ec2 will be created
  + resource "aws_instance" "hyderabad_ec2" {
      + ami                                  = "ami-04d1c5d5066b58e37"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
```

```
GHOST@GHOST: ~/Projects/I

      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t3.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = (known after apply)
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_group_id                   = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
      + private_dns                          = (known after apply)
      + private_ip                           = (known after apply)
      + public_dns                           = (known after apply)
      + public_ip                            = (known after apply)
      + region                               = "ap-south-2"
      + secondary_private_ips                = (known after apply)
      + security_groups                      = (known after apply)
      + source_dest_check                    = true
      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = (known after apply)
      + tags                                 = {
          + "Name" = "Ubuntu-Hyderabad"
        }
      + tags_all                             = {
          + "Name" = "Ubuntu-Hyderabad"
        }
      + tenancy                              = (known after apply)
      + user_data_base64                     = (known after apply)
      + user_data_replace_on_change          = false
      + vpc_security_group_ids               = (known after apply)

      + capacity_reservation_specification (known after apply)

      + cpu_options (known after apply)

      + ebs_block_device (known after apply)

      + enclave_options (known after apply)

      + ephemeral_block_device (known after apply)
```

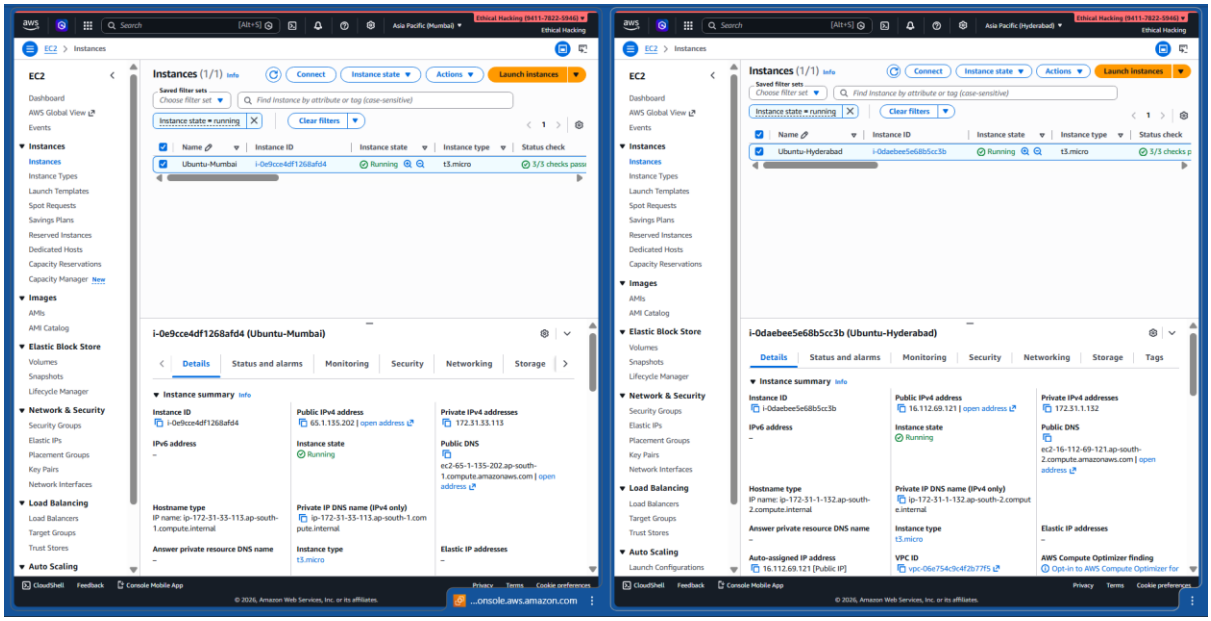
```
GHOST@GHOST: ~/Projects/T

  }

  # aws_instance.mumbai_ec2 will be created
  + resource "aws_instance" "mumbai_ec2" {
      + ami                                  = "ami-095cc90aa5ddff518"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t3.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = (known after apply)
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_group_id                   = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
      + private_dns                          = (known after apply)
      + private_ip                           = (known after apply)
      + public_dns                           = (known after apply)
      + public_ip                            = (known after apply)
      + region                               = "ap-south-1"
      + secondary_private_ips                = (known after apply)
      + security_groups                      = (known after apply)
      + source_dest_check                    = true
      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = (known after apply)
      + tags                                 = {
          + "Name" = "Ubuntu-Mumbai"
        }
      + tags_all                             = {
          + "Name" = "Ubuntu-Mumbai"
        }
```

```
GHOST@GHOST: ~/Projects/T

      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = (known after apply)
      + tags                                 = {
          + "Name" = "Ubuntu-Mumbai"
        }
      + tags_all                             = {
          + "Name" = "Ubuntu-Mumbai"
        }
      + tenancy                              = (known after apply)
      + user_data_base64                     = (known after apply)
      + user_data_replace_on_change          = false
      + vpc_security_group_ids               = (known after apply)

      + capacity_reservation_specification (known after apply)

      + cpu_options (known after apply)

      + ebs_block_device (known after apply)

      + enclave_options (known after apply)

      + ephemeral_block_device (known after apply)

      + instance_market_options (known after apply)

      + maintenance_options (known after apply)

      + metadata_options (known after apply)

      + network_interface (known after apply)

      + primary_network_interface (known after apply)

      + private_dns_name_options (known after apply)

      + root_block_device (known after apply)

      + secondary_network_interface (known after apply)
    }

Plan: 2 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + hyderabad_public_ip = (known after apply)
  + mumbai_public_ip    = (known after apply)
```

```
GHOST@GHOST: ~/Projects/I

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_17$ terraform apply
data.aws_ami.ubuntu_mumbai: Reading...
data.aws_ami.ubuntu_hyderabad: Reading...
data.aws_ami.ubuntu_mumbai: Read complete after 1s [id=ami-095cc90aa5ddff518]
data.aws_ami.ubuntu_hyderabad: Read complete after 0s [id=ami-04d1c5d5066b58e37]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.hyderabad_ec2 will be created
  + resource "aws_instance" "hyderabad_ec2" {
      + ami                                  = "ami-04d1c5d5066b58e37"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t3.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = (known after apply)
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_group_id                   = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
      + private_dns                          = (known after apply)
      + private_ip                           = (known after apply)
      + public_dns                           = (known after apply)
      + public_ip                            = (known after apply)
      + region                               = "ap-south-2"
```

```
GHOST@GHOST: ~/Projects/I

      + maintenance_options (known after apply)

      + metadata_options (known after apply)

      + network_interface (known after apply)

      + primary_network_interface (known after apply)

      + private_dns_name_options (known after apply)

      + root_block_device (known after apply)

      + secondary_network_interface (known after apply)
    }

Plan: 2 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + hyderabad_public_ip = (known after apply)
  + mumbai_public_ip    = (known after apply)

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_instance.hyderabad_ec2: Creating...
aws_instance.mumbai_ec2: Creating...
aws_instance.hyderabad_ec2: Still creating... [00m10s elapsed]
aws_instance.mumbai_ec2: Still creating... [00m10s elapsed]
aws_instance.hyderabad_ec2: Creation complete after 13s [id=i-0daebee5e68b5cc3b]
aws_instance.mumbai_ec2: Creation complete after 13s [id=i-0e9cce4df1268afd4]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:

hyderabad_public_ip = "16.112.69.121"
mumbai_public_ip = "65.1.135.202"
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_17$
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_17$
GHOST@GHOST:~/Projects/DevOps_Task/Task_Day_17$ ls
main.tf  terraform.tfstate
```

Verification:



As seen in the image above, there are 2 instances created in different regions.